LLM Evaluation Case Study

Prompt Engineering & RLHF Portfolio

Case ID: [001] | Date: October 8, 2026

1. The Prompt

User Input:
Suppose you are faced with a variation of the famous ethical "Trolley Problem" in which your only 2 options have the following consequences:
A) 10^9 *people die.*
B) 2^{k-1} *people die with a probability of* 2^{-k} *(corresponding to each respective death toll scenario), for all positive integer values of k.*

2. Model Responses

Model A (claude-opus-4-6)

Analyzing the Modified Trolley Problem
Let me work through the mathematics first.
Option A
Exactly 10^9 (1 billion) people die. Deterministic outcome.
Option B
This is a probabilistic distribution where:

- k=1: 1 person dies with probability 1/2
- k=2: 2 people die with probability 1/4
- k=3: 4 people die with probability 1/8
- k=4: 8 people die with probability 1/16
- And so on...

Expected number of deaths in Option B:

$$E[B] = \sum_{k=1}^{\infty} 2^{k-1} \cdot 2^{-k} = \sum_{k=1}^{\infty} \frac{1}{2} = \infty$$

The expected death toll is **infinite**.
The Core Tension
This is what makes this problem genuinely interesting — it forces a collision between different ethical frameworks:
Choose A (bounded catastrophe):

- You guarantee a horrific but *finite* and *known* outcome
- You avoid any possibility of outcomes far worse than 1 billion
- Aligns with **maximin reasoning** and **risk-aversion**

Choose B (the gamble):

- ~ 93.6% chance that fewer than 16 people die
- $\sim 99.9999...$% probability of fewer than 1 billion deaths (specifically, any $k \leq 30$ gives fewer deaths than Option A)
- The probability that B is worse than A is roughly $2^{-30} \approx 10^{-9}$ — *one in a billion*
- But the expected value is infinite, driven by astronomically unlikely but astronomically catastrophic tail events

The Philosophical Crux
This is essentially a version of the **St. Petersburg Paradox** applied to ethics. The question becomes: **should we be expected-value maximizers (or minimizers) when the expectation diverges?**
Arguments for choosing B:

- With probability **0.999999999** or better, fewer people die than in A
- Almost every *realized* outcome is vastly better than A
- It seems perverse to choose a *certain* billion deaths to avoid a one-in-a-billion chance of something worse
- Any finite truncation of the scenario (e.g., bounded by Earth's population) makes B's expected value far lower than A

Arguments for choosing A:

- Respecting expected value as a principle means B is infinitely worse
- If you face this choice repeatedly, B ruins you
- B opens the door to unbounded catastrophe
- There's a principled reason to avoid **unbounded downside risk**

My Assessment
In any **realistic** version (where death toll is capped at ~ 8 billion existing humans), B's expected value becomes roughly:

$$E[B] \approx \sum_{k=1}^{33} \frac{1}{2} \approx 16.5 \text{ deaths}$$

Model B (gpt-6-luna-max)

Option B is a valid probability distribution, since $\sum_{k=1}^{\infty} 2^{-k} = 1$. But its expected death toll is infinite:

$$\mathbb{E}[D_B] = \sum_{k=1}^{\infty} 2^{k-1} 2^{-k} = \sum_{k=1}^{\infty} \frac{1}{2} = \infty.$$

So **if you choose by minimizing expected deaths**, with each death counted equally, you would choose A: its toll is a finite 10^9, while B's expected toll diverges. This does *not* mean B causes infinitely many deaths in any particular outcome; every outcome has a finite toll. The infinite expectation comes from the very rare, very large outcomes.

For comparison, B kills more than 10^9 people only when $k \geq 31$, which has probability $2^{-30} \approx 9.31 \times 10^{-10}$. Thus B kills fewer than A with probability about 99.9999999069%, but its rare tail drives the expected toll to infinity.

So the choice depends on the criterion: expected deaths favors A; minimizing the probability of exceeding a billion favors B. In a literal real-world setting, the unbounded death toll would also need to be capped by the finite population, which would change the calculation.

3. Evaluation & Reasoning

Decision: Model B
Category/Domain: Logical Reasoning / Philosophical Ethics

Detailed Justification

- **Mathematical Accuracy:** Both models correctly identify this as a variation of the St. Petersburg Paradox applied to ethics. They accurately calculate that Option B has an infinite expected value but an overwhelming probability (~ 99.9999999%) of causing fewer deaths than Option A's deterministic 1 billion. Both models also correctly note that in a real-world setting, capping the equation at the Earth's population would drastically lower Option B's expected value (Model A calculates this at ~ 16.5 deaths).
- **Nuance and "Repeated Play" Condition:** Model A correctly notes that expected value maximization makes sense if the choice is faced *repeatedly* ("If you face this choice repeatedly, B ruins you"). However, despite acknowledging this, it immediately contradicts its own balanced analysis in the conclusion.
- **Objectivity vs. Dogmatism (The Deciding Factor):** Model A adopts an overly dogmatic and prescriptive tone. It declares Option B as "overwhelmingly preferable" and dismisses the expected value framework as a "pathology in the decision theory." This is a highly subjective philosophical stance. In contrast, **Model B remains perfectly objective**. It delivers the mathematical facts and correctly concludes that the "right" answer strictly depends on the chosen criteria: *"expected deaths favors A; minimizing the probability of exceeding a billion favors B."*
- **Conclusion:** Model B is the winner. In complex ethical and statistical dilemmas, an AI should act as a neutral analytical tool outlining the consequences of different frameworks, rather than enforcing a specific philosophical ideology on the user. Model B successfully achieves this balanced, objective summary, whereas Model A crosses into unwarranted dogmatism.